Service Memo

Date: 12/26/2023 Memo #:

Subject: OCCT Post-Build Burn-In & Stability Testing

Why:

The first Gear at Micro Center is to Take Care of the Customer. In our Service Centers, we execute on this Gear by providing Best in Class service to Customer PCs and Laptops or units that will eventually end up in our customer's hands.

We are rolling out the OCCT stability testing software to be used during the post-build burn-in and stability testing to ensure customers receive their systems with no errors or hardware issues. The testing procedure provided is to be run on all custom builds to ensure system stability.

How:

Section A – Stability Testing

1. Insert a flash drive containing the **start-occt.bat** script and the **OCCT** folder.
2. Double click **start-occt.bat** in the root folder.
3. If presented with a Windows security prompt, click **Yes** to allow the program to run.
4. OCCT will copy itself and all dependencies to the current user desktop and continue to run.
    a. It is now safe to remove your flash drive.
5. After 30 seconds, OCCT will automatically run through the four tests outlined in the "Explanation of Stability Testing Structure" section below, without the need for any interaction.
    a. If you press **any** key before those 30 seconds, it will take you to an individual component testing menu. Continue to Section B for more information.
6. When the stability testing schedule has completed, the results will be output to the **Micro Center QA Results** folder on the current user desktop.
7. The results from the current desktop will be copied to the Default User desktop, and the Default User desktop will open in a new window to confirm the files are present.

Section B – Diagnostics and Individual Component Testing

1. In the individual component testing menu, select Option 1, 2, 3 or 4 depending on which specific test set you wish to run.
2. Once complete, results will be saved to the current Desktop in a folder named **OCCT Diagnostics**.
3. If needed, review the reports in the **OCCT Diagnostics** folder, then delete the folder.

Note: To re-run OCCT stability testing after you have already run it once, open the OCCT folder on the current user Desktop, and double click **run-occt.bat**. It is not necessary to re-run from your USB drive unless you have deleted the OCCT folder from the desktop.

Results and Output

The burn in results will be saved to the **Micro Center QA Results** folder on the current user profile Desktop with the filename **Burn In Results** in a CSV, JSON, and HTML format. The HTML file is a formatted webpage in which you can view all results of each piece of the stability test, as well as all monitoring metrics such as temperature, power consumption, clock speeds, etc.

Full Details:

File Tree

- start-occt.bat
- OCCT (Folder)
    - OCCT Usage Memo.pdf
    - run-occt.bat
    - occt-2.bat
    - resources (Folder)
        - OCCTEnterprise.exe
        - license.oke
        - configs (Folder)
            - 2min-power.json
            - 10min-cpu.json
            - 10min-gpu.json
            - 10min-mem.json
            - 30m-burn-in-json

Explanation of Stability Testing Structure

1. **Power Supply (2 Minutes)**
   Hardware Tested: CPU, GPU, PSU
   Test: CPU/GPU are run at 100% utilization to ensure PSU will provide clean and constant power as configured.
   Failure Identification: System will reboot or shut down. No error message will be present on screen.
2. **CPU Power Stress (10 Minutes)**
   Hardware Tested: CPU, CPU Cooler, Memory Controller, Memory
   Test: Large Data Set hammers CPU to both ensure stability and error checking, while also testing the cooling capacity of the installed CPU cooler.
   Failure Identification: Red indicator will appear in top left corner, and errors will be present in status window.
3. **Memory Error Check (10 Minutes)**
   Hardware Tested: Memory, Memory Controller
   Test: 80% of usable memory is run through both SSE and AVX2 instruction sets to check for errors.
   Failure Identification: Red indicator will appear in top left corner, and errors will be present in status window.
4. **3D Adaptive GPU Stress (10 Minutes)**
   Hardware Tested: GPU
   Test: GPU 3D processor is tested using variable settings ranging from 10% to 100% intensity.
   Failure Identification: Red indicator will appear in top left corner, and errors will be present in status window.

Screenshots and References

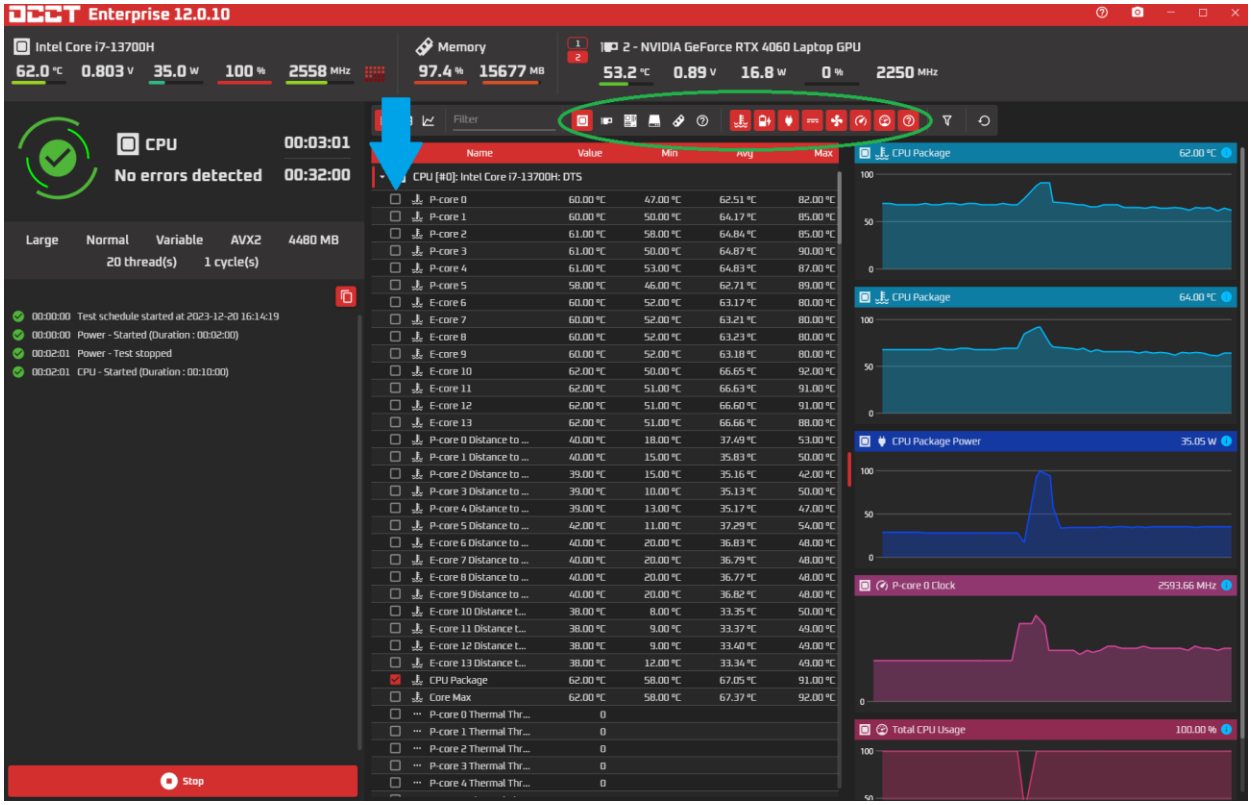


Figure 1 – The OCCT Testing Interface when running a predefined OCCT stability test. The two sections on the right side of the interface show all current system information readings. In the green circle, click any square to toggle that filter on or off. Right-click any filter to toggle solo mode, which will turn off all other filters. Click the checkboxes marked with the blue arrow to determine which graphs are present on the right side.

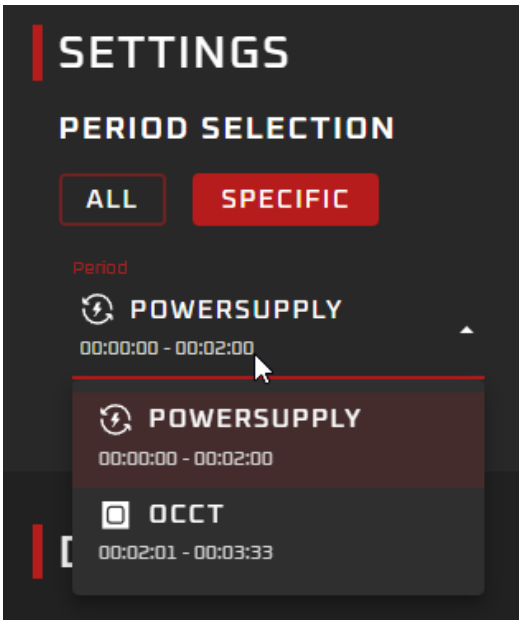


Figure 2 – In the Burn In Results file, you can view the full test results, or each specific section of the testing, such as CPU testing, GPU testing, etc.

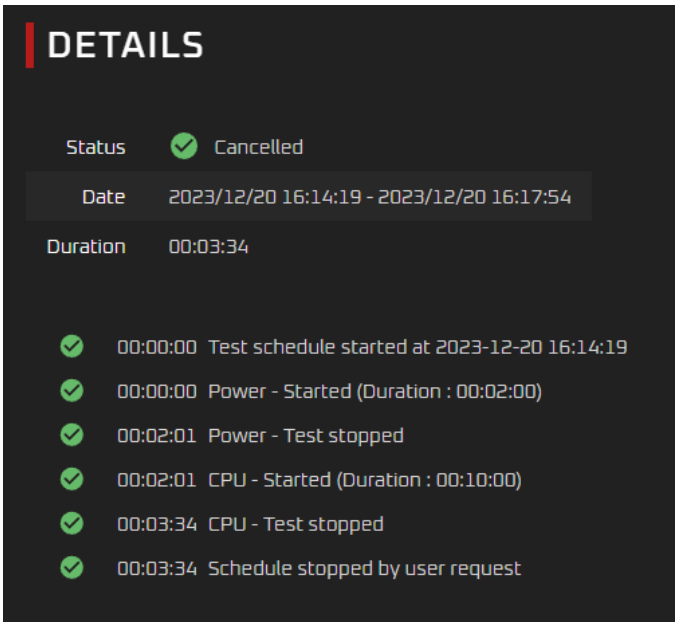


Figure 3 – The Burn In Results file will give the total test duration, and a breakdown of each of the piece of the test.

SYSTEM INFORMATION

CPU | MOTHERBOARD | MEMORY | GPU

GENERAL

INTEL CORE I7-13700H

Name	Intel Core i7-13700H		
CPU Count	1		
Cores (Physical / Logical)	14 / 20		
Features	Hyperthreading - AVX - AVX2		
Temperature	↓ 8 °C	÷ 50.38 °C	↑ 92 °C
Frequency	↓ 1.34 MHz	÷ 1589.46 MHz	↑ 4328.50 MHz
FSB	↓ 99.76 MHz	÷ 99.76 MHz	↑ 99.78 MHz
Multiplier	↓ 4x	÷ 17.08x	↑ 45x
Power	↓ 17.34 W	÷ 34.13 W	↑ 115.02 W

SYSTEM INFORMATION

CPU | MOTHERBOARD | MEMORY | GPU

GENERAL

MEMORY STICK #0

MEMORY STICK #1

Type	DDR5		
Channels Active / Supported	4 / 4		
Total size	16384 MB		
Frequency	↓ 2194.63 MHz	÷ 2281.99 MHz	↑ 2394.14 MHz

TIMINGS

SYSTEM INFORMATION

CPU | MOTHERBOARD | MEMORY | GPU

GENERAL

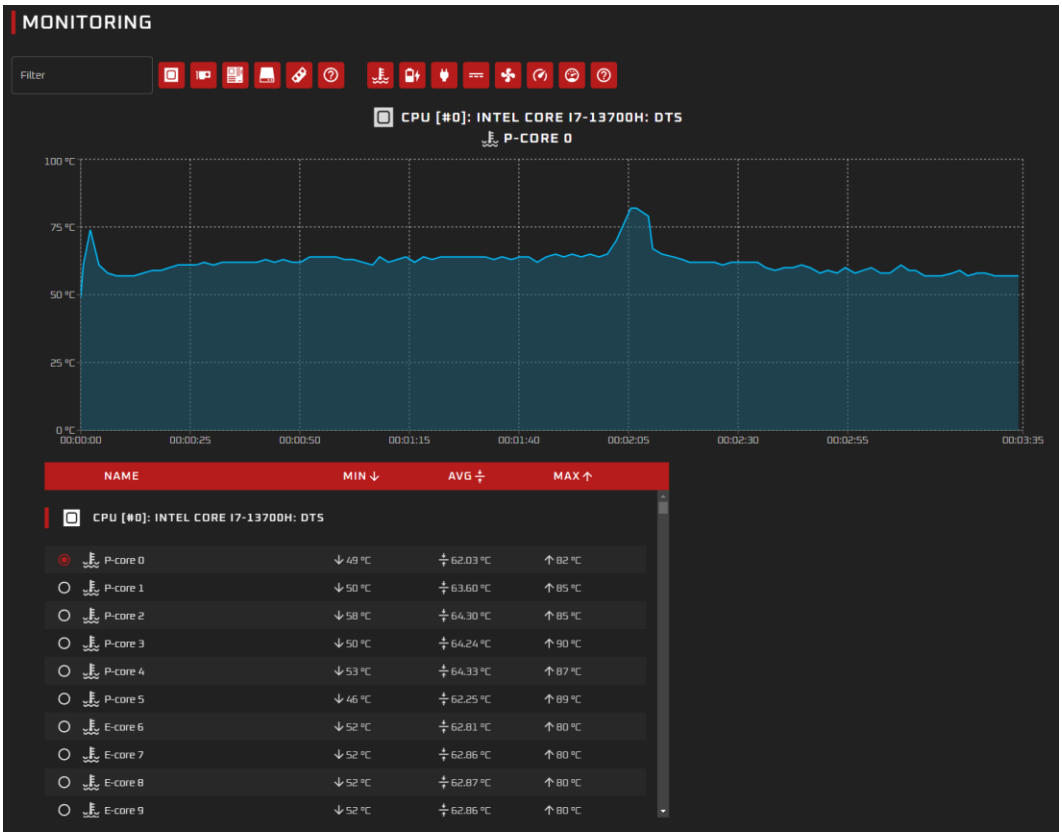
MEMORY STICK #0

MEMORY STICK #1

Name	Row: 0 [BANK 0/Controller0-ChannelA-DIMM0] - 8 GB PC5-38400 DDR5 SDRAM SK Hynix HMCG66MEBSA095N
Manufacturer	SK Hynix
Chip manufacturer	SK Hynix
Base speed	2400.0 MHz (DDR5-4800 / PC5-38400)
Part Number	HMCG66MEBSA095N
Serial Number	3138931904 (C04C18BB)
ECC	No

DETAILS

Figure 4, 5 & 6 – The Burn In Results will provide a breakdown of all system information, including serial numbers, timings, temperatures, and drive/firmware versions where applicable.



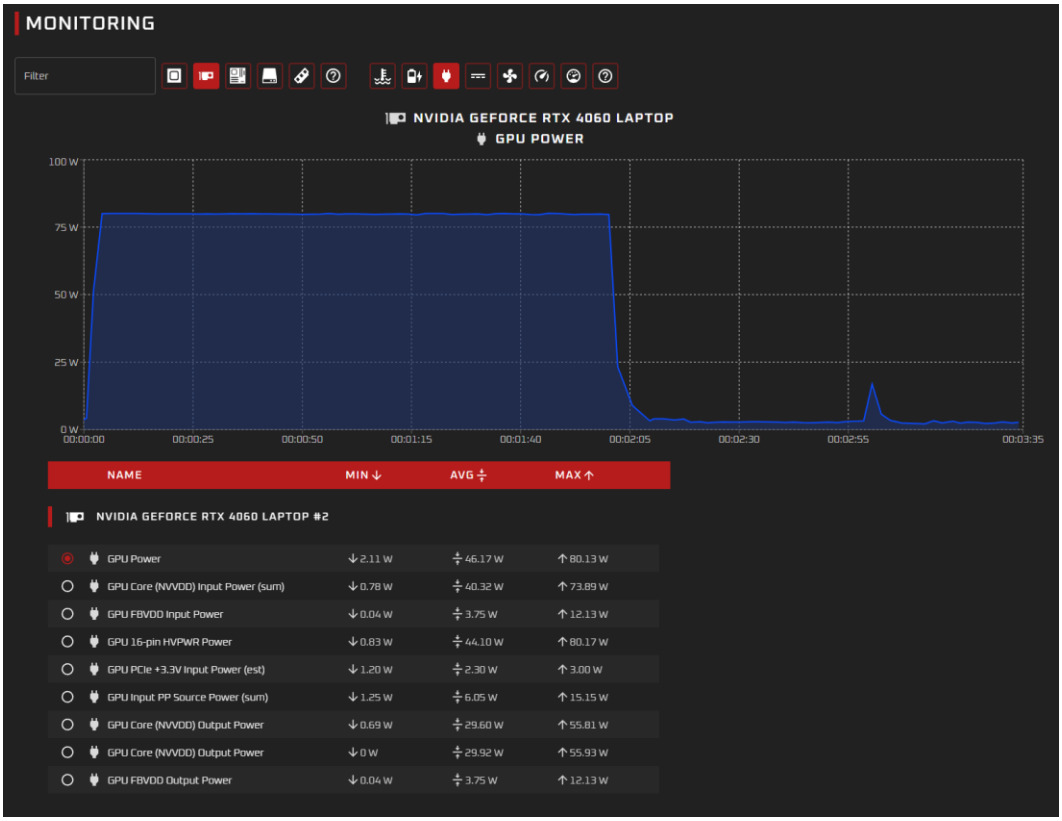


Figure 7 & 8 – At the bottom of the Burn In Results report, an interactive graph is present showing all sensor readings over time. Click the buttons at the top can toggle sensors on/off, similar to the OCCT testing interface.

Store Responsibility:

- Execute stability testing using the procedure above
- Review testing output to ensure no errors are present

Home Office Responsibility:

- Provide OCCT software and/or script updates as needed

Customer Responsibility:

- None